

Operating Liquidity Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Operating Liquidity Coverage (OLC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on OLC achieves an annualized gross (net) Sharpe ratio of 0.51 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 35 (29) bps/month with a t-statistic of 3.30 (2.72), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance) is 34 bps/month with a t-statistic of 3.10.

1 Introduction

Production-based asset pricing theory posits that firm characteristics reflecting production decisions and investment frictions explain cross-sectional variation in expected returns through their connection to firms' marginal costs of production and exposure to aggregate productivity shocks. While extensive empirical evidence documents the predictive power of accounting-based measures for stock returns, the precise economic mechanisms linking firms' operational characteristics to risk premia remain incompletely understood. The ability of firms to cover short-term obligations through operating activities represents a fundamental aspect of production efficiency, yet the asset pricing implications of operating liquidity management have received limited attention in the production-based framework.

Operating Liquidity Coverage (OLC) captures firms' capacity to meet immediate obligations through cash generation from operations, directly reflecting the efficiency of the production process and the firm's exposure to productivity shocks. In a production economy with capital adjustment costs, firms with lower operating liquidity coverage face higher marginal costs of production, as constrained internal cash generation necessitates more costly external financing for investment [Gomes \(2001\)](#); [Zhang \(2005\)](#). These financing frictions amplify the firm's conditional risk premium through increased exposure to aggregate productivity shocks, as documented in models where financial constraints interact with production decisions [Livdan et al. \(2009\)](#). Furthermore, firms with inadequate operating liquidity coverage exhibit greater sensitivity to investment irreversibility and adjustment costs, mechanisms that [Cooper and Priestley \(2011\)](#) demonstrate generate substantial cross-sectional variation in expected returns. The production-based framework predicts that low OLC firms command higher expected returns as compensation for their elevated exposure to systematic productivity risk. This prediction aligns with the broader literature linking operational efficiency measures to risk premia through the investment channel

[Liu et al. \(2009\)](#), where firms facing binding constraints in converting operations to cash flows require higher returns to compensate investors for bearing additional systematic risk.

We find that a value-weighted long/short trading strategy based on Operating Liquidity Coverage achieves an annualized gross Sharpe ratio of 0.51, with the strategy earning average monthly returns of 42 basis points (t-statistic = 3.87). The strategy delivers economically and statistically significant abnormal returns across all standard factor models, with monthly alphas ranging from 27 to 35 basis points relative to the Fama-French three-, four-, five-, and six-factor models, all with t-statistics exceeding 2.63. The predictive power of OLC remains robust after accounting for transaction costs, with the net Sharpe ratio of 0.47 placing the strategy in the top 2% of documented anomalies. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLC strategy achieves an average return of 38 basis points per month with a t-statistic of 2.98, demonstrating that the effect is not confined to small, illiquid stocks. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the OLC trading strategy maintains an alpha of 34 basis points per month with a t-statistic of 3.10, confirming that OLC captures distinct information about expected returns beyond existing predictors.

This paper contributes to the production-based asset pricing literature by establishing Operating Liquidity Coverage as a robust cross-sectional predictor that directly maps to theoretical mechanisms of production efficiency and investment frictions. Our findings extend [Zhang \(2005\)](#) by demonstrating that operational cash generation capacity, beyond traditional investment measures, captures firms' exposure to productivity shocks and generates significant risk premia. Unlike [Lamont et al. \(2001\)](#), who focus on financial constraints through external financing measures, we show that internal liquidity generation from operations provides distinct

information about conditional risk premia in the production economy. Relative to [Balakrishnan et al. \(2010\)](#), who examine broader liquidity implications for asset pricing, our focus on operating liquidity coverage specifically ties cash generation efficiency to production-based risk factors. Methodologically, we demonstrate that OLC improves the mean-variance efficient frontier even after controlling for established factors and closely related anomalies, with the strategy maintaining statistical significance across alternative portfolio construction methods and after accounting for transaction costs. The economic magnitude and statistical robustness of the OLC effect suggest that operational efficiency measures warrant greater attention in production-based asset pricing models, particularly as they relate to firms’ ability to navigate productivity shocks and investment frictions through internally generated resources.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. COMPUSTAT offers standardized financial data that enables consistent measurement across firms and over time. Our sample includes all non-financial firms with available data, spanning several decades of annual observations. Operating Liquidity Coverage signal utilizes two key accounting variables from firms’ financial statements. Cash (CH) represents the total cash and cash equivalents reported on a firm’s balance sheet, including currency, demand deposits, and highly liquid short-term investments that can be readily converted to known amounts of cash. Operating income after depreciation (OIADP) measures a firm’s operating performance by capturing the income generated from core business operations after accounting for depreciation expense but before deducting interest and taxes. This measure reflects the cash-

generating ability of a firm’s operating activities while incorporating the non-cash depreciation charge. We construct the Operating Liquidity Coverage signal by dividing cash (CH) by operating income after depreciation (OIADP) for each firm-year observation. This ratio captures the relationship between a firm’s liquid assets and its operating profitability, essentially measuring how many years of current operating income the firm’s cash holdings represent. Higher values indicate that a firm maintains substantial cash reserves relative to its operating earnings, which may reflect either conservative financial management or potential operational challenges generating income from assets. We calculate this measure annually using fiscal year-end values and require non-missing values for both cash and operating income after depreciation. To ensure meaningful ratios, we exclude observations where operating income after depreciation equals zero or is negative, as these cases would produce undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OLC signal. Panel A plots the time-series of the mean, median, and interquartile range for OLC. On average, the cross-sectional mean (median) OLC is 1.04 (0.32) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input OLC data. The signal’s interquartile range spans -1.19 to 2.60. Panel B of Figure 1 plots the time-series of the coverage of the OLC signal for the CRSP universe. On average, the OLC signal is available for 6.60% of CRSP names, which on average make up 7.09% of total market capitalization.

4 Does OLC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OLC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OLC portfolio and sells the low OLC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short OLC strategy earns an average return of 0.42% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.27% to 0.35% per month and have t-statistics exceeding 2.63 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.16, with a t-statistic of 6.38 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 461 stocks and an average market capitalization of at least \$1,141 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 30 bps/month with a t-statistics of 1.82. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 22-42bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OLC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the OLC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OLC, as well as average returns and alphas for long/short trading OLC strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLC strategy achieves an average return of 38 bps/month with a t-statistic of 2.98. Among these large cap stocks, the alphas for the OLC strategy relative to the five most common factor models range from 24 to 44 bps/month with t-statistics between 1.98 and 3.51.

5 How does OLC perform relative to the zoo?

Figure 2 puts the performance of OLC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OLC strategy falls in the distribution. The OLC strategy’s gross (net) Sharpe ratio of 0.51 (0.47) is greater than 90% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OLC strategy (red line).² Ignoring trading costs, a \$1 invested in the OLC strategy would have yielded \$11.28 which ranks the OLC strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OLC strategy would have yielded \$9.00 which ranks the OLC strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the OLC relative to those. Panel A shows that the OLC strategy gross alphas fall between the 60 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OLC strategy has a positive net generalized alpha for five out of the five factor models. In these cases OLC ranks between the 77 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OLC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of OLC with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OLC or at least to weaken the power OLC has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of OLC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLC}OLC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OLC. Stocks are finally grouped into five OLC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OLC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OLC signal in these Fama-MacBeth regressions exceed -0.91, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on OLC is -0.58.

Similarly, Table 5 reports results from spanning tests that regress returns to the OLC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OLC strategy earns alphas that range from 32-35bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.99, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OLC trading strategy achieves an alpha of 34bps/month with a t-statistic of 3.10.

7 Does OLC add relative to the whole zoo?

Finally, we can ask how much adding OLC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the OLC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes OLC grows to \$2286.59.

8 Conclusion

This study provides compelling evidence that Operating Liquidity Coverage (OLC) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OLC is available.

by Novy-Marx and Velikov (2023), demonstrates that OLC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and implementation costs.

The key findings of our research underscore the economic importance of OLC as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.51 gross (0.47 net of transaction costs), indicating strong performance that remains viable in practical implementation. Most notably, the strategy delivers statistically significant abnormal returns of 35 basis points per month (t -statistic = 3.30) relative to the Fama-French five-factor model augmented with momentum, with these returns remaining economically meaningful at 29 basis points after accounting for transaction costs. The robustness of our results is further validated by the persistence of significant alpha (34 bps/month, t -statistic = 3.10) even when controlling for six closely related strategies from the factor zoo, including profitability, value, and various volatility measures.

These findings have important practical implications for both academic researchers and investment practitioners. For academics, OLC emerges as a distinct source of cross-sectional return predictability that is not fully captured by existing factor models or related anomalies. For practitioners, the strategy’s resilience to transaction costs and its orthogonality to established factors suggest potential portfolio diversification benefits and alpha generation opportunities in quantitative investment strategies.

Despite these promising results, several limitations warrant consideration. First, while our analysis accounts for standard transaction costs, the actual implementation costs may vary depending on market conditions, trade size, and execution capabilities. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings to different market regimes or international markets. Third, while OLC demonstrates statistical significance, the economic mechanisms

driving this predictability require further theoretical development and empirical investigation.

Future research should explore several promising avenues. Investigating the performance of OLC across different market conditions, including periods of financial stress and varying liquidity regimes, would provide valuable insights into the strategy's robustness. Additionally, examining the international evidence for OLC's predictive power could reveal whether this anomaly reflects a global phenomenon or is specific to certain market structures. Researchers should also investigate the underlying economic channels through which operating liquidity coverage affects firm valuation and expected returns, potentially linking it to theories of financial constraints, operational efficiency, or investor attention. Finally, exploring the interaction between OLC and corporate events such as earnings announcements, credit rating changes, or macroeconomic shocks could enhance our understanding of when and why this signal is most effective in predicting returns.

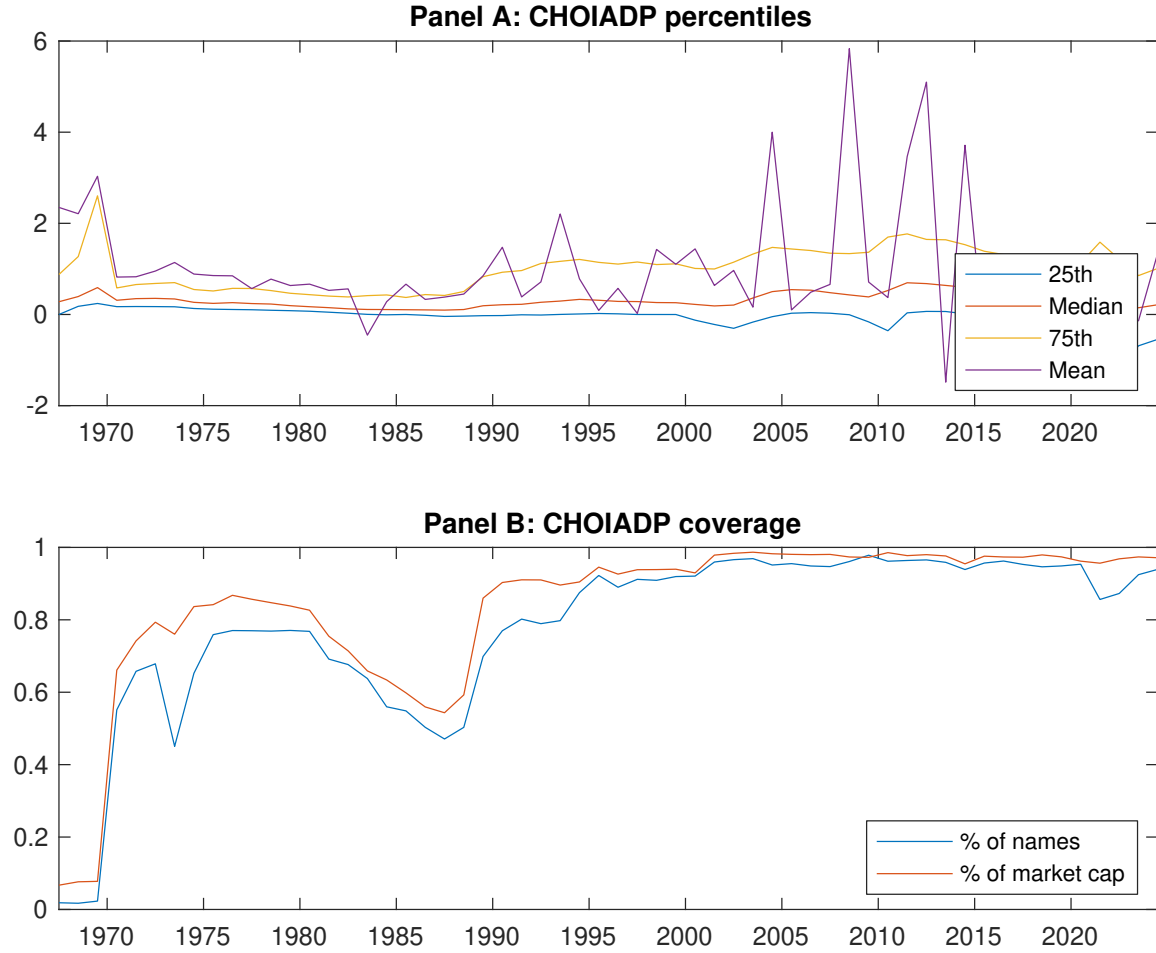


Figure 1: Times series of OLC percentiles and coverage. This figure plots descriptive statistics for OLC. Panel A shows cross-sectional percentiles of OLC over the sample. Panel B plots the monthly coverage of OLC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OLC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on OLC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37 [1.98]	0.58 [3.25]	0.48 [2.54]	0.50 [2.40]	0.78 [3.66]	0.42 [3.87]
α_{CAPM}	-0.21 [-2.82]	0.04 [0.49]	-0.11 [-1.47]	-0.15 [-1.94]	0.11 [1.48]	0.31 [3.01]
α_{FF3}	-0.19 [-2.59]	0.03 [0.34]	-0.14 [-1.85]	-0.16 [-2.02]	0.08 [1.22]	0.27 [2.63]
α_{FF4}	-0.19 [-2.62]	0.02 [0.24]	-0.14 [-1.87]	-0.13 [-1.69]	0.12 [1.77]	0.32 [3.03]
α_{FF5}	-0.10 [-1.39]	-0.10 [-1.38]	-0.23 [-3.14]	-0.17 [-2.13]	0.22 [3.20]	0.32 [2.98]
α_{FF6}	-0.11 [-1.53]	-0.10 [-1.35]	-0.23 [-3.07]	-0.15 [-1.86]	0.24 [3.54]	0.35 [3.30]
Panel B: Fama and French (2018) 6-factor model loadings for OLC-sorted portfolios						
β_{MKT}	0.94 [53.93]	0.98 [55.93]	1.04 [60.00]	1.09 [57.42]	1.11 [69.16]	0.16 [6.38]
β_{SMB}	0.03 [1.32]	-0.06 [-2.23]	0.03 [1.23]	0.08 [2.87]	0.08 [3.52]	0.05 [1.31]
β_{HML}	-0.00 [-0.05]	0.00 [0.05]	0.04 [1.20]	0.01 [0.21]	0.08 [2.55]	0.08 [1.65]
β_{RMW}	-0.19 [-5.58]	0.32 [9.16]	0.24 [6.92]	0.07 [2.01]	-0.30 [-9.58]	-0.11 [-2.19]
β_{CMA}	-0.10 [-1.96]	0.08 [1.57]	0.06 [1.17]	-0.03 [-0.58]	-0.09 [-1.98]	0.01 [0.11]
β_{UMD}	0.02 [0.98]	-0.00 [-0.11]	-0.00 [-0.19]	-0.03 [-1.55]	-0.04 [-2.37]	-0.05 [-2.18]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1120	461	520	622	924	
me (\$10 ⁶)	1141	2537	2907	2235	1805	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OLC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.42 [3.87]	0.31 [3.01]	0.27 [2.63]	0.32 [3.03]	0.32 [2.98]	0.35 [3.30]
Quintile	NYSE	EW	0.42 [3.06]	0.50 [3.72]	0.40 [3.25]	0.31 [2.47]	0.17 [1.46]	0.10 [0.88]
Quintile	Name	VW	0.30 [1.82]	0.38 [2.35]	0.23 [1.52]	0.22 [1.46]	-0.05 [-0.37]	-0.04 [-0.28]
Quintile	Cap	VW	0.38 [3.65]	0.28 [2.78]	0.28 [2.78]	0.32 [3.16]	0.39 [3.85]	0.42 [4.07]
Decile	NYSE	VW	0.46 [3.28]	0.47 [3.32]	0.37 [2.69]	0.40 [2.91]	0.27 [1.94]	0.30 [2.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.39 [3.59]	0.27 [2.61]	0.23 [2.27]	0.26 [2.52]	0.26 [2.47]	0.29 [2.72]
Quintile	NYSE	EW	0.22 [1.61]	0.31 [2.21]	0.20 [1.60]	0.15 [1.22]		
Quintile	Name	VW	0.25 [1.53]	0.32 [1.94]	0.17 [1.13]	0.16 [1.10]		
Quintile	Cap	VW	0.35 [3.36]	0.24 [2.37]	0.24 [2.36]	0.26 [2.60]	0.32 [3.13]	0.34 [3.33]
Decile	NYSE	VW	0.42 [2.96]	0.42 [2.95]	0.32 [2.35]	0.34 [2.49]	0.22 [1.60]	0.26 [1.86]

Table 3: Conditional sort on size and OLC

This table presents results for conditional double sorts on size and OLC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OLC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OLC and short stocks with low OLC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OLC Quintiles					OLC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.37 [1.14]	0.33 [1.09]	0.83 [3.12]	0.79 [3.26]	0.90 [3.64]	0.52 [2.97]	0.68 [3.96]	0.53 [3.35]	0.50 [3.11]	0.19 [1.27]	0.18 [1.23]
	(2)	0.46 [1.55]	0.59 [2.38]	0.85 [3.61]	0.76 [3.26]	0.66 [2.57]	0.32 [1.97]	0.43 [2.68]	0.27 [1.79]	0.25 [1.67]	0.07 [0.52]	0.07 [0.46]
	(3)	0.52 [2.04]	0.40 [1.72]	0.83 [3.68]	0.69 [2.89]	0.71 [3.15]	0.16 [1.21]	0.21 [1.61]	0.10 [0.75]	-0.01 [-0.08]	-0.09 [-0.74]	-0.17 [-1.37]
	(4)	0.41 [1.87]	0.61 [3.00]	0.48 [2.10]	0.77 [3.51]	0.85 [3.66]	0.44 [3.48]	0.38 [3.02]	0.35 [2.75]	0.31 [2.39]	0.34 [2.63]	0.31 [2.36]
	(5)	0.36 [2.05]	0.57 [3.14]	0.35 [1.77]	0.48 [2.42]	0.74 [3.47]	0.38 [2.98]	0.25 [2.02]	0.24 [1.98]	0.31 [2.46]	0.39 [3.19]	0.44 [3.51]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OLC Quintiles					OLC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	416	407	421	423	424	33	29	40	43	40	
	(2)	117	113	117	118	113	63	62	66	66	63	
	(3)	80	79	83	80	78	107	107	116	109	105	
	(4)	63	64	64	64	63	216	226	226	226	227	
(5)	56	56	56	56	56	1183	2104	2007	1647	1531		

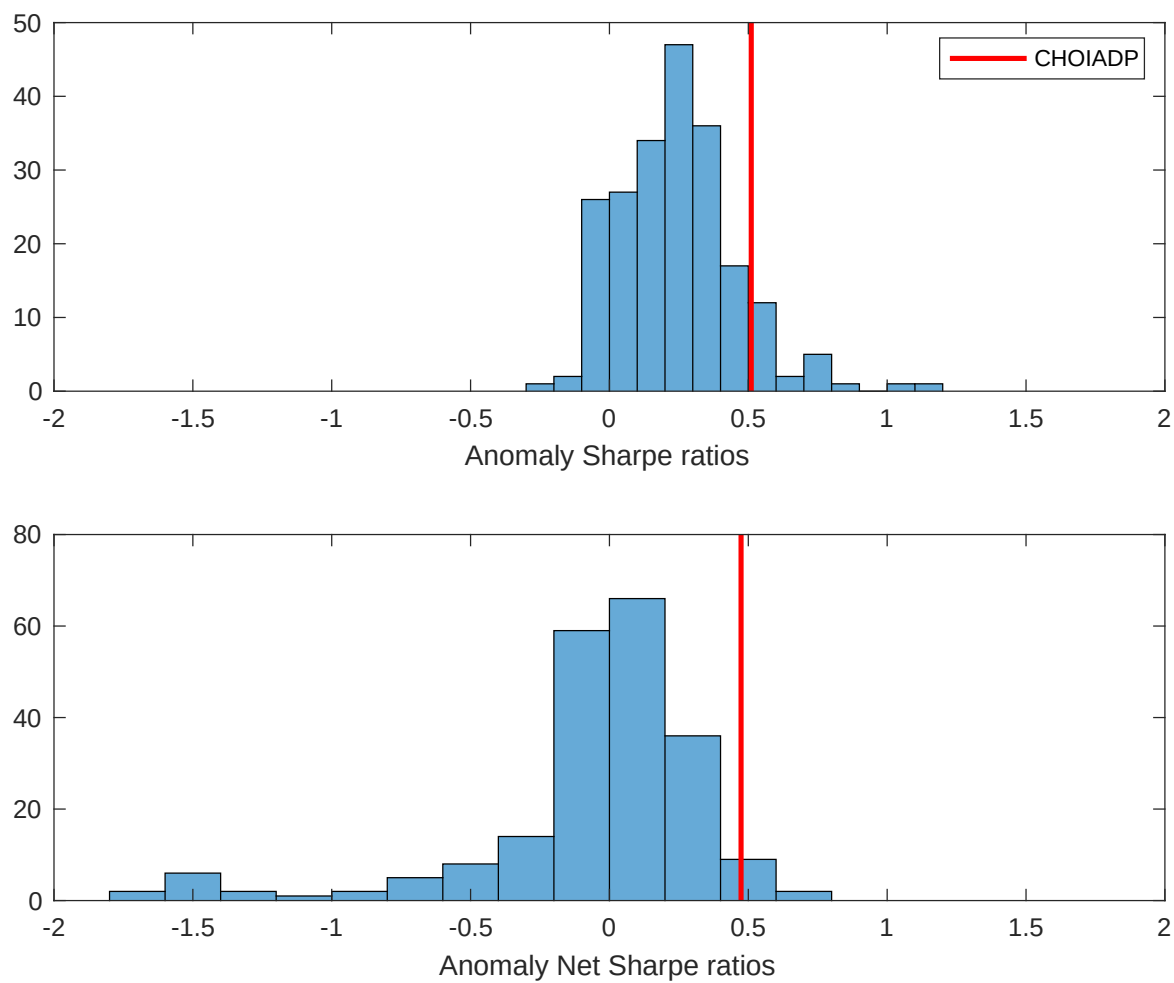


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OLC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

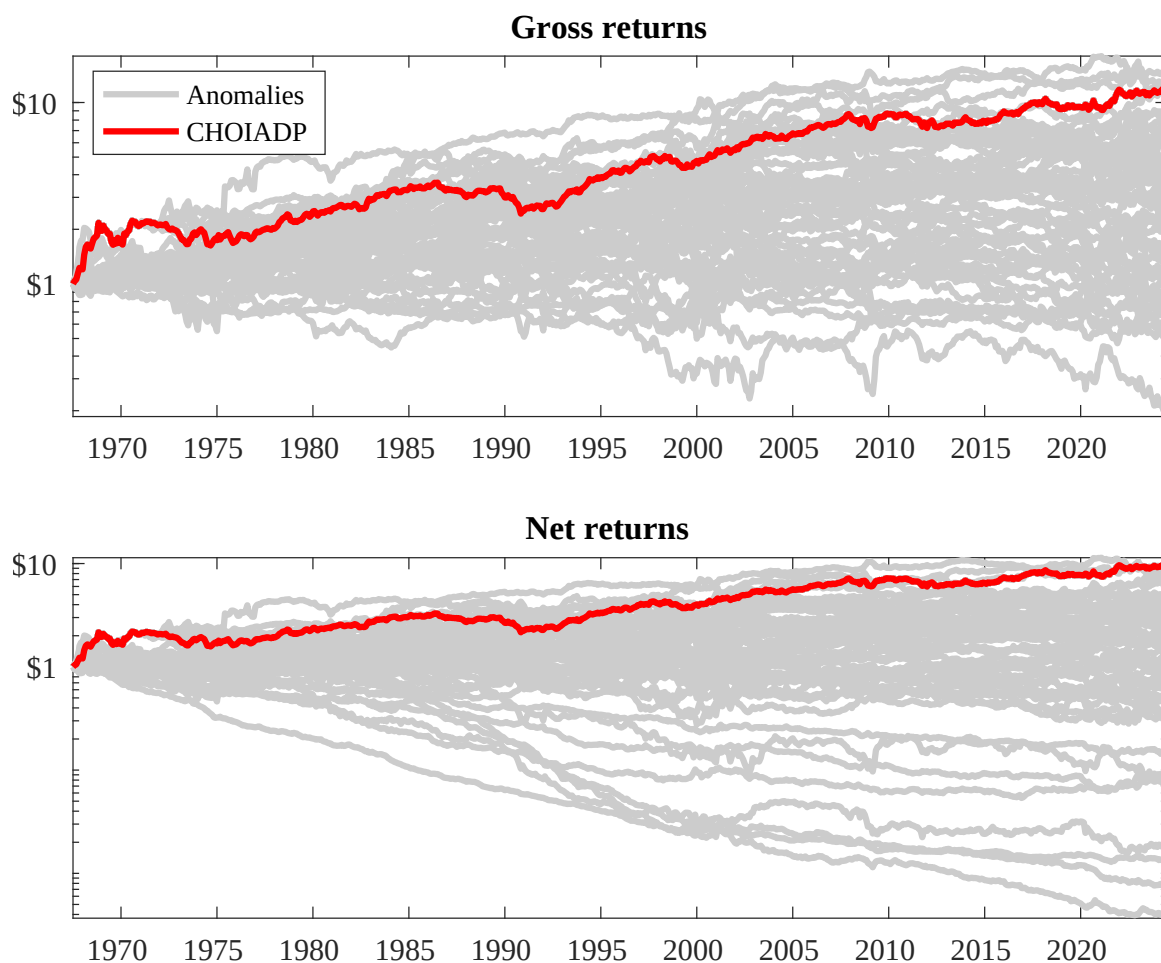
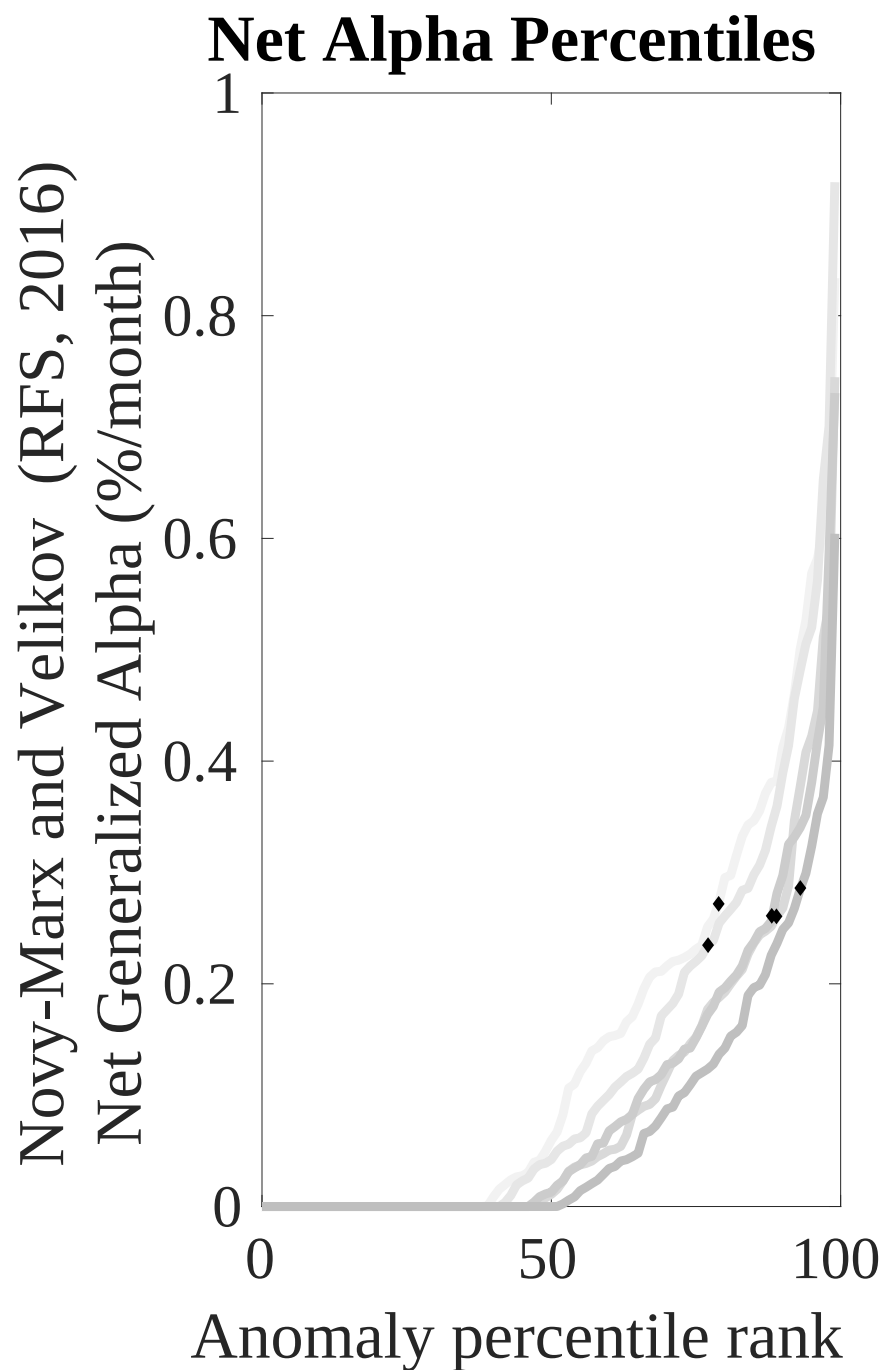
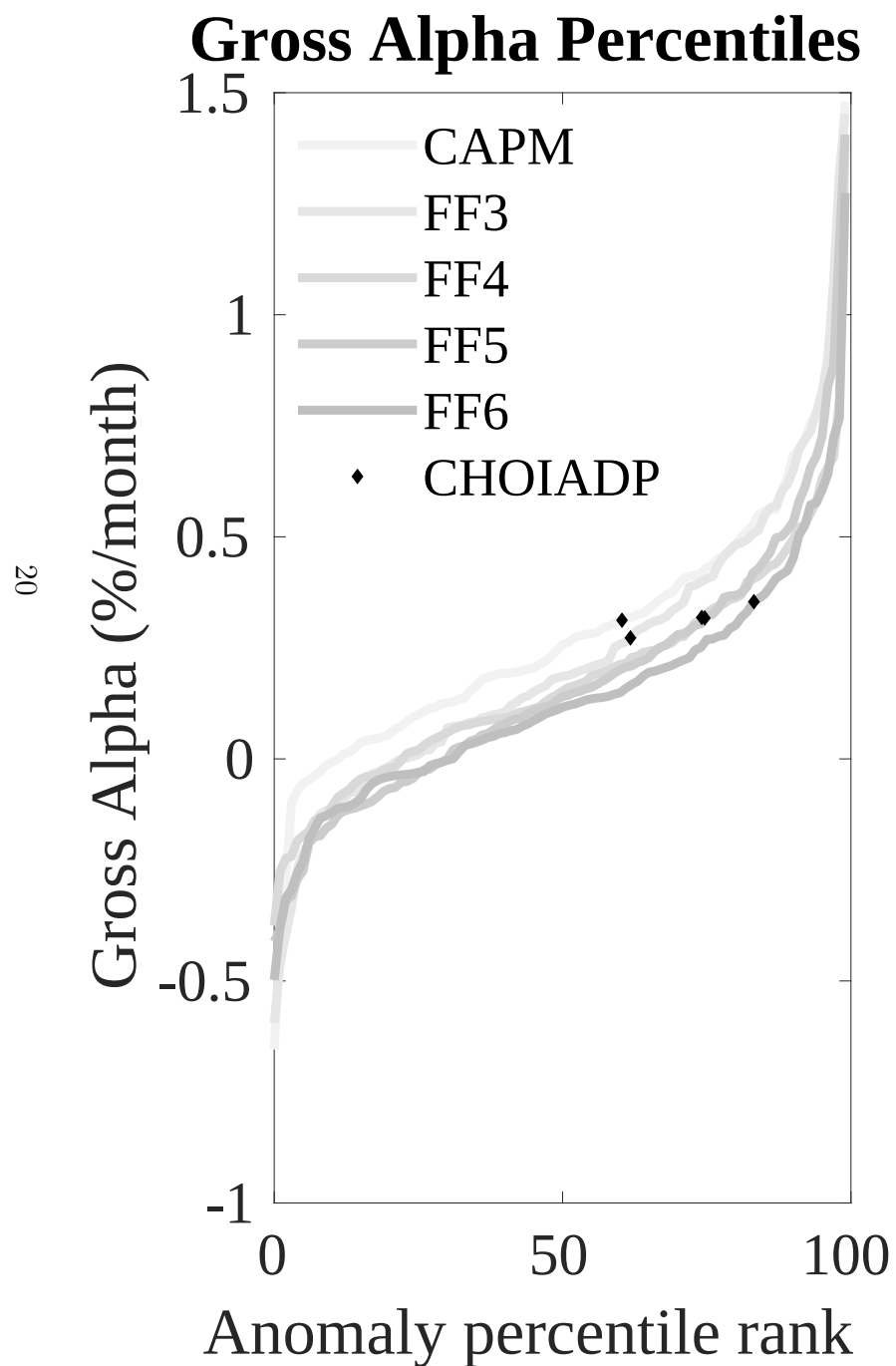


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OLC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



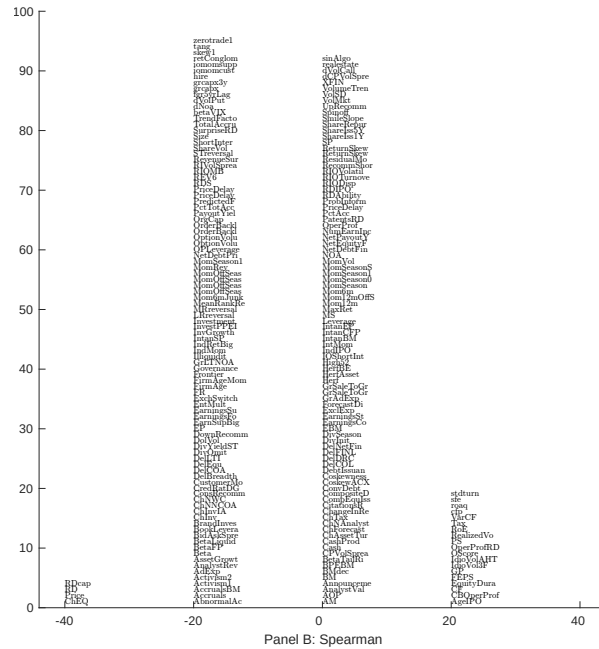
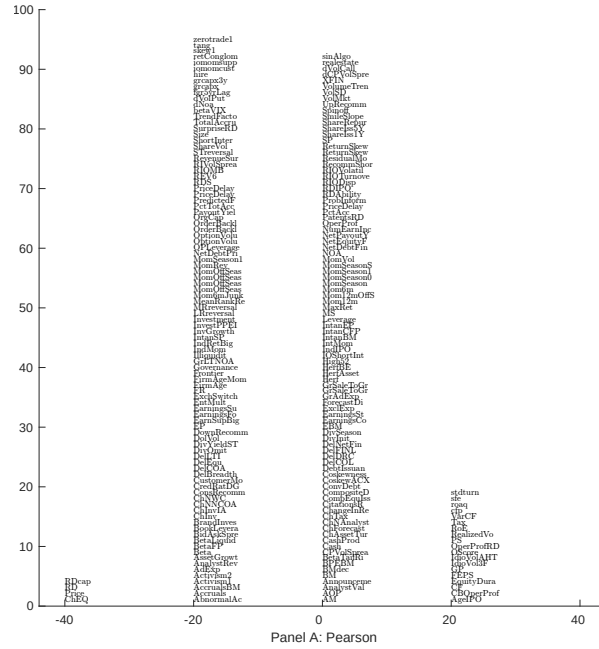


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with OLC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

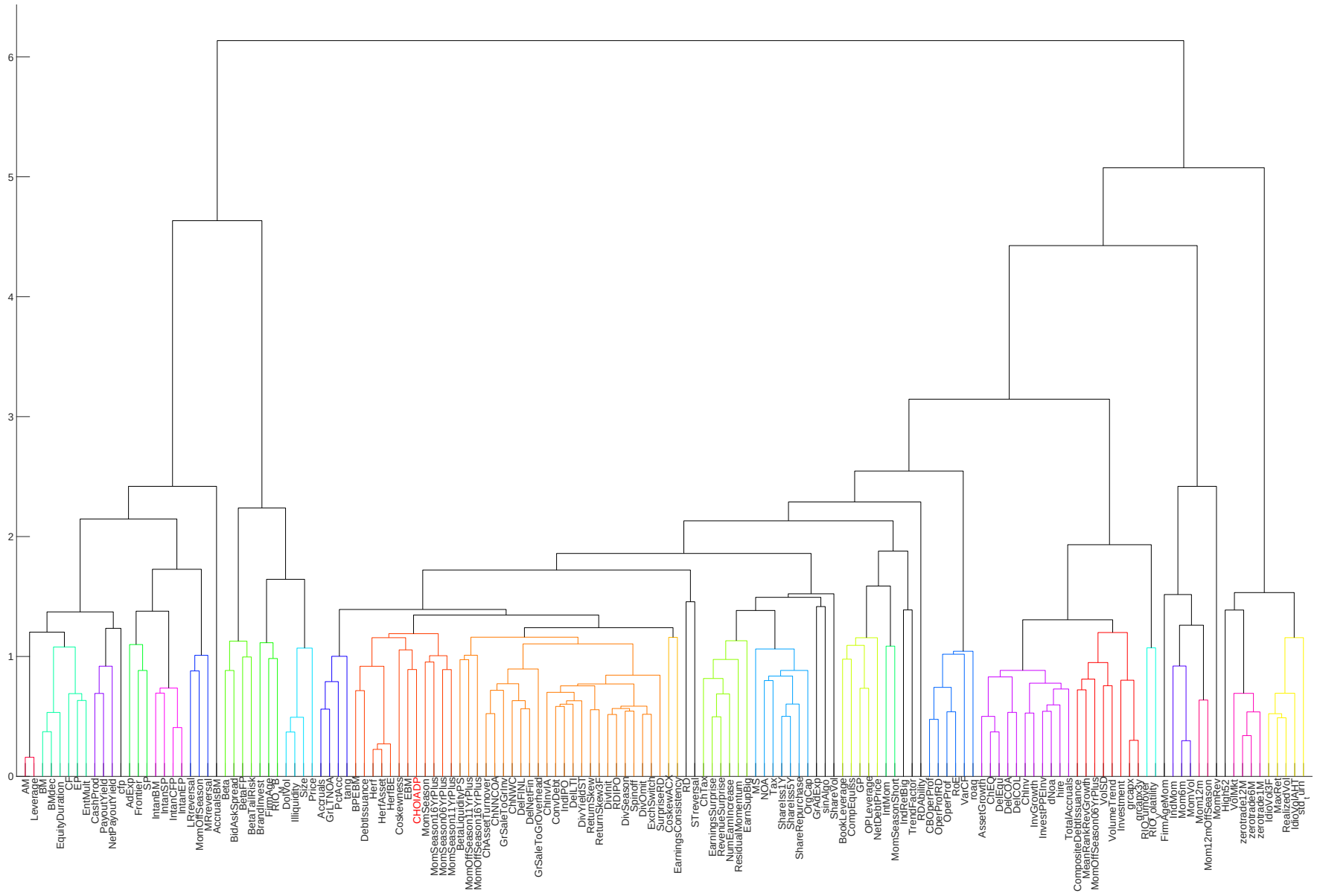


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

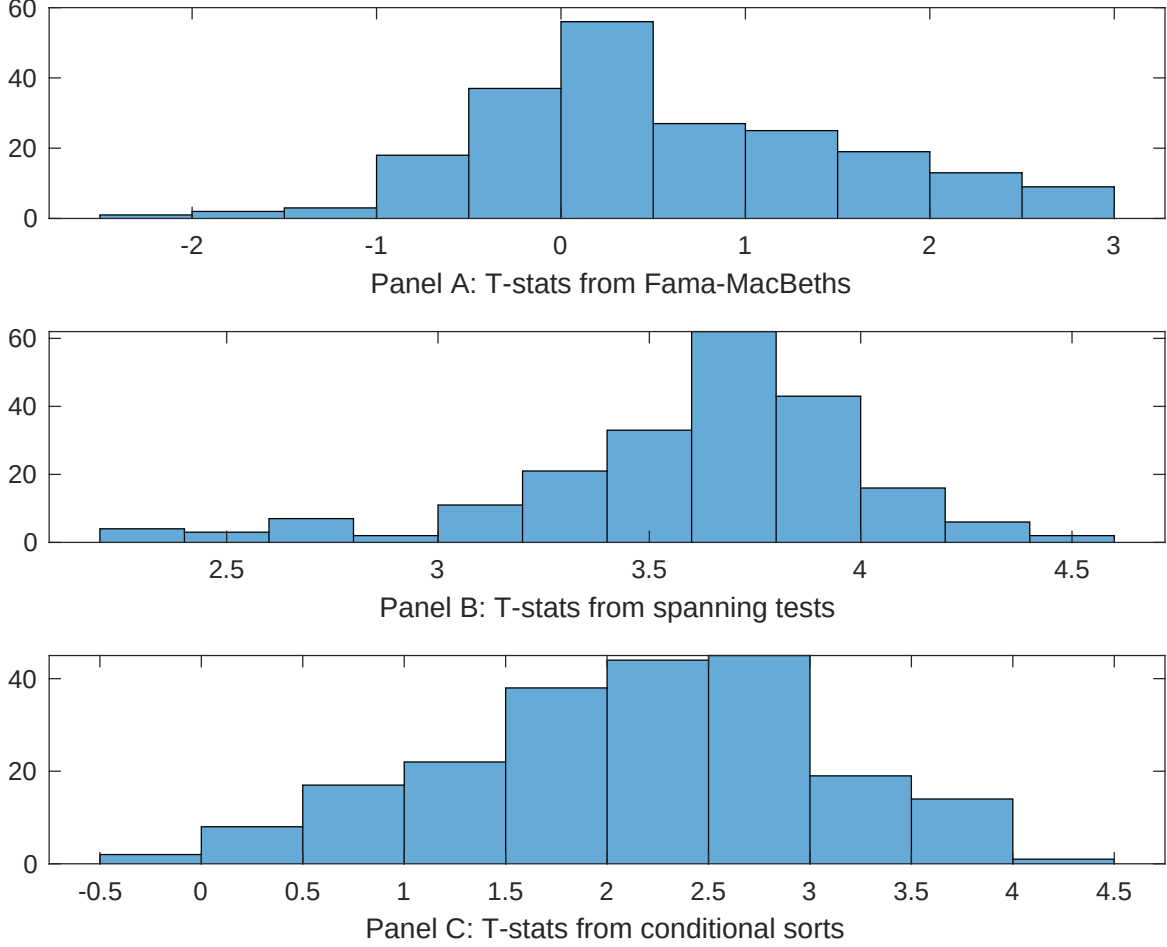


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OLC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLC} OLC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OLC. Stocks are finally grouped into five OLC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OLC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OLC} OLC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.12 [4.95]	0.12 [4.28]	0.15 [8.03]	0.14 [7.51]	0.15 [8.73]	0.12 [5.48]	0.17 [8.79]
OLC	-0.58 [-0.08]	0.53 [0.08]	-0.49 [-0.82]	-0.45 [-0.84]	-0.53 [-0.91]	0.37 [0.52]	-0.31 [-0.58]
Anomaly 1	0.43 [0.03]						0.12 [0.15]
Anomaly 2		0.61 [1.28]					0.90 [3.21]
Anomaly 3			0.16 [3.58]				-0.12 [-1.05]
Anomaly 4				0.96 [1.71]			-0.20 [-0.45]
Anomaly 5					0.15 [3.55]		0.28 [2.53]
Anomaly 6						-0.94 [-0.82]	-0.13 [-2.70]
# months	648	648	643	643	643	643	643
$\bar{R}^2(\%)$	0	1	2	2	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OLC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OLC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are net income / book equity, Price, Idiosyncratic risk (3 factor), Idiosyncratic risk (AHT), Realized (Total) Volatility, Cash-flow to price variance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.35 [3.23]	0.35 [3.22]	0.33 [3.04]	0.32 [2.99]	0.33 [3.05]	0.33 [3.02]	0.34 [3.10]
Anomaly 1	-14.39 [-2.38]						-18.46 [-2.78]
Anomaly 2		-6.49 [-1.45]					-4.31 [-0.84]
Anomaly 3			0.06 [0.02]				-5.81 [-0.77]
Anomaly 4				5.20 [1.40]			11.03 [1.81]
Anomaly 5					0.90 [0.28]		-1.34 [-0.23]
Anomaly 6						2.42 [0.58]	3.04 [0.61]
mkt	14.29 [5.35]	17.40 [6.65]	16.45 [5.77]	18.44 [6.34]	16.86 [5.70]	17.14 [6.09]	16.90 [5.53]
smb	0.79 [0.19]	12.27 [2.03]	5.68 [1.15]	10.75 [2.06]	6.28 [1.43]	7.35 [1.54]	11.46 [1.79]
hml	7.44 [1.50]	9.91 [2.00]	8.85 [1.75]	7.37 [1.46]	8.57 [1.70]	9.92 [1.89]	7.74 [1.32]
rmw	2.15 [0.29]	-14.66 [-2.66]	-11.22 [-1.97]	-16.04 [-2.65]	-11.91 [-2.12]	-11.98 [-2.32]	-2.49 [-0.31]
cma	-2.65 [-0.36]	0.84 [0.11]	0.73 [0.10]	-1.31 [-0.18]	0.30 [0.04]	1.32 [0.18]	-3.82 [-0.48]
umd	-4.80 [-1.91]	-8.86 [-2.47]	-5.13 [-1.95]	-5.97 [-2.30]	-5.31 [-2.03]	-5.48 [-2.11]	-8.00 [-2.19]
# months	684	684	680	680	680	680	680
$\bar{R}^2(\%)$	12	11	11	11	11	11	12

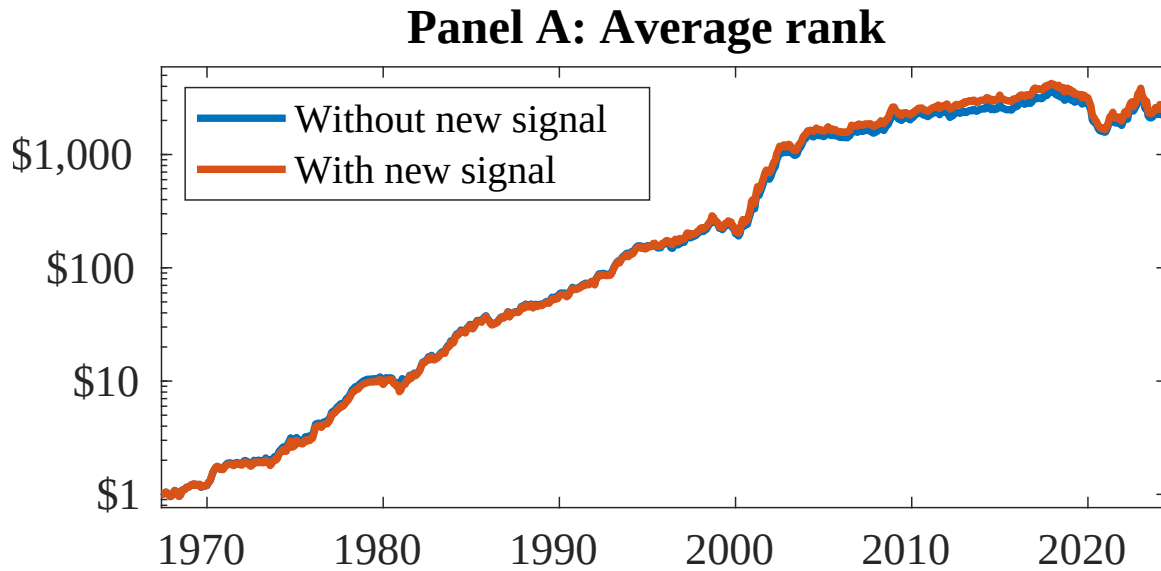


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as OLC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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